

Lecture 09 - Class Activity - Correlation Linear Regression

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In class activity 9: Correlation and Linear Regression

Introduction

This document demonstrates key concepts in correlation and regression analysis using ecological examples, focusing on:

1. Understanding correlation vs. regression
2. Calculating and interpreting correlation coefficients
3. Testing correlation assumptions
4. Performing simple linear regression
5. Checking regression assumptions
6. Interpreting regression output and ANOVA tables

We'll work with real ecological datasets to practice these concepts.

Part 1: Load Required Packages and Data

```
# Load required packages
library(tidyverse) # For data manipulation and visualization
library(patchwork) # For combining plots
library(car)       # For regression diagnostics
library(broom)     # For tidy model output

# Set seed for reproducible results
set.seed(123)

# Create the datasets from the lecture
# Lion data from Example 17.1
l_df <- tibble(
  proportion_black = c(0.21, 0.14, 0.11, 0.13, 0.12, 0.13, 0.12, 0.18, 0.23, 0.22,
                      0.20, 0.17, 0.15, 0.27, 0.26, 0.21, 0.30, 0.42, 0.43, 0.59,
                      0.60, 0.72, 0.29, 0.10, 0.48, 0.44, 0.34, 0.37, 0.34, 0.74, 0.79,
                      0.51),
  age_years = c(1.1, 1.5, 1.9, 2.2, 2.6, 3.2, 3.2, 2.9, 2.4, 2.1,
               1.9, 1.9, 1.9, 1.9, 2.8, 3.6, 4.3, 3.8, 4.2, 5.4,
               5.8, 6.0, 3.4, 4.0, 7.3, 7.3, 7.8, 7.1, 7.1, 13.1, 8.8, 5.4)
)

# Booby data from Example 16.1
b_df <- tibble(
  visits_as_nestling = c(1, 7, 15, 4, 11, 14, 23, 14, 9, 5, 4, 10,
                        13, 13, 14, 12, 13, 9, 8, 18, 22, 22, 23, 31),
  future_aggression = c(-0.80, -0.92, -0.80, -0.46, -0.47, -0.46, -0.23, -0.16,
                        -0.23, -0.23, -0.16, -0.10, -0.10, 0.04, 0.13, 0.19,
                        0.25, 0.23, 0.15, 0.23, 0.31, 0.18, 0.17, 0.39)
)

# Prairie stability data from Example 17.3
```

```
p_df <- tibble(
  species_number = rep(c(1, 2, 4, 8, 16), times = c(32, 32, 32, 32, 33)),
  log_stability = 1.20 + 0.033 * species_number + rnorm(161, 0, 0.35)
)
```

Package Overview

- **tidyverse**: Collection of packages for data science
- **patchwork**: Combine multiple ggplot2 plots easily
- **car**: Companion to Applied Regression (diagnostic tools)
- **broom**: Convert statistical objects into tidy data frames

Part 2: Correlation Analysis

Correlation Analysis: Data Types and Assumptions

Data Types Required:

- **X variable**: Continuous numerical
- **Y variable**: Continuous numerical - Both variables should be measured (not manipulated)

Assumptions for Pearson Correlation:

- Random sampling from the population
- Bivariate normality (both variables normally distributed)
- Linear relationship between variables
- No extreme outliers

Calculating Correlation Coefficients

Let's start with the Nazca booby data to explore correlation:

```
# Calculate Pearson correlation coefficient
booby_corr <- cor(b_df$visits_as_nestling, b_df$future_aggression)
booby_corr
```

```
[1] 0.5337225
```

```
# Perform correlation test
booby_cor_test <- cor.test(b_df$visits_as_nestling, b_df$future_aggression)

booby_cor_test
```

Pearson's product-moment correlation

```
data: b_df$visits_as_nestling and b_df$future_aggression
t = 2.9603, df = 22, p-value = 0.007229
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1660840 0.7710999
```

```
sample estimates:
      cor
0.5337225
```

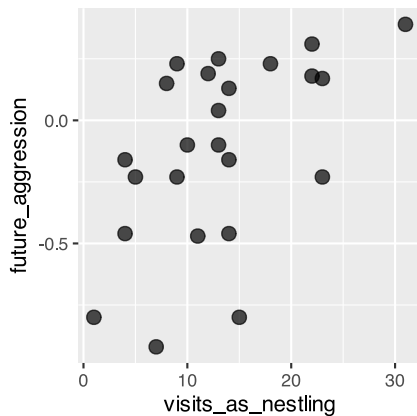
```
# Calculate R-squared (variance explained)
r_squared <- booby_corr^2
r_squared
```

```
[1] 0.2848597
```

Visualizing the Correlation

```
# Create scatterplot with correlation
booby_plot <- ggplot(b_df, aes(x = visits_as_nestling, y = future_aggression)) +
  geom_point(size = 3, alpha = 0.7)
# geom_smooth(method = "lm", se = TRUE, color = "blue", alpha = 0.2) +

booby_plot
```



! Activity 1: Interpret the Correlation

Based on the output above, answer these questions:

1. **Direction:** Is the correlation positive or negative? What does this mean biologically?
 - Your answer: _____
2. **Strength:** How would you classify this correlation (weak, moderate, strong)?
 - Your answer: _____
3. **Significance:** Is the correlation statistically significant? What is the p-value?
 - Your answer: _____
4. **Variance explained:** What percentage of variance in adult aggression is explained by nestling visits?
 - Your answer: _____

Testing Correlation Assumptions

```
# Test normality of each variable
shapiro_visits <- shapiro.test(b_df$visits_as_nestling)
shapiro_aggression <- shapiro.test(b_df$future_aggression)
shapiro_visits
```

Shapiro-Wilk normality test

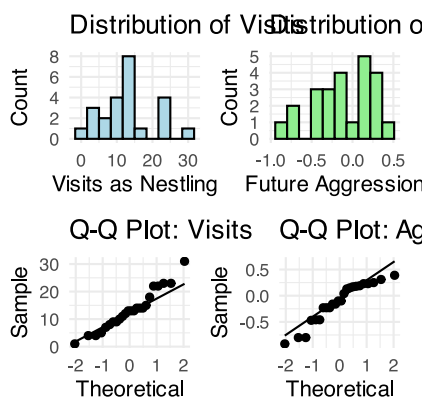
```
data: b_df$visits_as_nestling  
W = 0.95783, p-value = 0.3965
```

shapiro_aggression

Shapiro-Wilk normality test

```
data: b_df$future_aggression  
W = 0.91575, p-value = 0.04709
```

```
# Create diagnostic plots  
p1 <- ggplot(b_df, aes(x = visits_as_nestling)) +  
  geom_histogram(bins = 10, fill = "lightblue", color = "black") +  
  labs(title = "Distribution of Visits", x = "Visits as Nestling", y = "Count") +  
  theme_minimal()  
  
p2 <- ggplot(b_df, aes(x = future_aggression)) +  
  geom_histogram(bins = 10, fill = "lightgreen", color = "black") +  
  labs(title = "Distribution of Aggression", x = "Future Aggression", y = "Count") +  
  theme_minimal()  
  
p3 <- ggplot(b_df, aes(sample = visits_as_nestling)) +  
  stat_qq() + stat_qq_line() +  
  labs(title = "Q-Q Plot: Visits", x = "Theoretical", y = "Sample") +  
  theme_minimal()  
  
p4 <- ggplot(b_df, aes(sample = future_aggression)) +  
  stat_qq() + stat_qq_line() +  
  labs(title = "Q-Q Plot: Aggression", x = "Theoretical", y = "Sample") +  
  theme_minimal()  
  
# Combine plots  
(p1 + p2) / (p3 + p4)
```



⚠ When Assumptions Are Violated

If normality assumptions are violated ($p < 0.05$ in Shapiro-Wilk test), consider:

1. **Spearman's rank correlation** (non-parametric alternative)
2. **Data transformation** (log, square root, etc.)
3. **Removing outliers** (if justified)

Let's try Spearman's correlation:

```
# Calculate Spearman's rank correlation
spearman_test <- cor.test(b_df$visits_as_nestling,
                          b_df$future_aggression,
                          method = "spearman")
```

```
Warning in cor.test.default(b_df$visits_as_nestling, b_df$future_aggression, :
Cannot compute exact p-value with ties
```

```
spearman_test
```

Spearman's rank correlation rho

```
data: b_df$visits_as_nestling and b_df$future_aggression
S = 1213.5, p-value = 0.01976
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.472374
```

```
# Compare with Pearson
print(paste("Pearson r:", round(booby_corr, 3)))
```

```
[1] "Pearson r: 0.534"
```

```
print(paste("Spearman rho:", round(spearman_test$estimate, 3)))
```

```
[1] "Spearman rho: 0.472"
```

Part 3: Simple Linear Regression

Now let's move from correlation to regression using the lion nose data.

i Linear Regression: Data Types and Assumptions

Data Types Required:

- **X variable (predictor):** Continuous numerical
- **Y variable (response):** Continuous numerical - X can be fixed/controlled, Y is the outcome of interest

Assumptions for Linear Regression:

- **Linearity:** Relationship between X and Y is linear
- **Independence:** Observations are independent
- **Homoscedasticity:** Constant variance of residuals
- **Normality:** Residuals are normally distributed
- **No influential outliers**

Fitting a Linear Regression Model

```
# Fit linear regression model
lion_model <- lm(age_years ~ proportion_black, data = l_df)

# Get model summary
summary(lion_model)
```

Call:

```
lm(formula = age_years ~ proportion_black, data = l_df)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.5449	-1.1117	-0.5285	0.9635	4.3421

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.8790	0.5688	1.545	0.133
proportion_black	10.6471	1.5095	7.053	7.68e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.669 on 30 degrees of freedom

Multiple R-squared: 0.6238, Adjusted R-squared: 0.6113

F-statistic: 49.75 on 1 and 30 DF, p-value: 7.677e-08

! Activity 2: Interpret the Regression Output

From the regression output above:

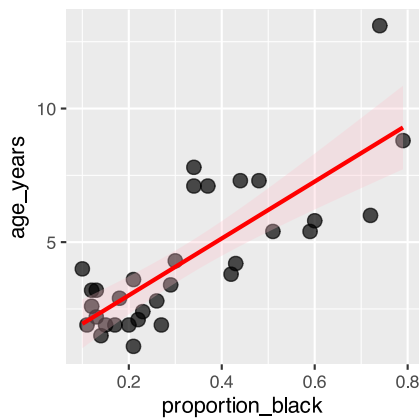
1. **Regression equation:** Write the equation in the form: $\text{age} = \text{_____} + \text{_____} \times \text{proportion_black}$
 - Your answer: _____
2. **Slope interpretation:** What does the slope value mean in biological terms?
 - Your answer: _____
3. **R-squared:** What percentage of variation in age is explained by nose blackness?
 - Your answer: _____
4. **Significance:** Is the relationship statistically significant? How do you know?
 - Your answer: _____

Visualizing the Regression

```
# Create regression plot with confidence interval
lion_plot <- ggplot(l_df, aes(x = proportion_black, y = age_years)) +
  geom_point(size = 3, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, color = "red", fill = "pink", alpha = 0.3)

lion_plot
```

``geom_smooth()`` using formula = 'y ~ x'



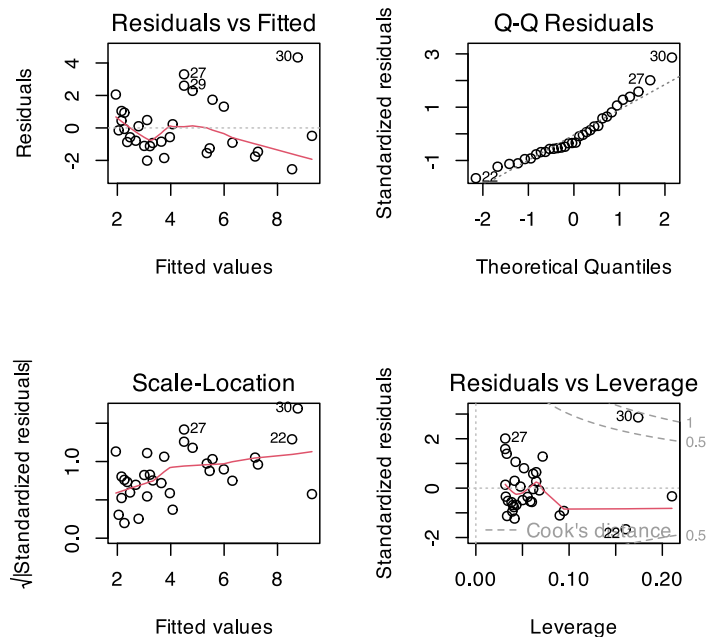
💡 Confidence vs. Prediction Intervals

- **Confidence Interval:** Range for the mean age of ALL lions with that nose blackness
- **Prediction Interval:** Range for an INDIVIDUAL lion with that nose blackness
- Prediction intervals are always wider than confidence intervals

Part 4: Testing Regression Assumptions

Diagnostic Plots

```
# Create diagnostic plots
par(mfrow = c(2, 2))
plot(lion_model)
```



```
par(mfrow = c(1, 1))
```

Interpreting Diagnostic Plots

i Understanding Regression Diagnostic Plots

1. **Residuals vs Fitted:**
 - Look for: Random scatter around horizontal line at 0
 - Problems: Patterns indicate non-linearity or heteroscedasticity
2. **Q-Q Plot:**
 - Look for: Points following the diagonal line
 - Problems: Deviations indicate non-normal residuals
3. **Scale-Location:**
 - Look for: Random scatter with horizontal trend line
 - Problems: Increasing spread indicates heteroscedasticity
4. **Residuals vs Leverage:**
 - Look for: Points within Cook's distance lines
 - Problems: Points outside indicate influential observations

Formal Tests of Assumptions


```
# Test for normality of residuals
shapiro_residuals <- shapiro.test(residuals(lion_model))
shapiro_residuals
```

Shapiro-Wilk normality test

```
data: residuals(lion_model)
W = 0.93879, p-value = 0.0692
```

```
# Test for homoscedasticity (Breusch-Pagan test)
library(lmtest)
bp_test <- bptest(lion_model)
bp_test
```

studentized Breusch-Pagan test

```
data: lion_model
BP = 6.8946, df = 1, p-value = 0.008646
```

! Activity 3: Assess Assumption Violations

Based on the diagnostic plots and tests:

1. **Linearity:** Does the relationship appear linear? (Check Residuals vs Fitted plot)
 - Your answer: _____
2. **Normality:** Are the residuals normally distributed? (Check Q-Q plot and Shapiro test)
 - Your answer: _____
3. **Homoscedasticity:** Is the variance constant? (Check Scale-Location plot and BP test)
 - Your answer: _____
4. **Influential points:** Are there any concerning influential observations?
 - Your answer: _____

Part 5: ANOVA for Regression

Understanding Variance Partitioning

```
# Get ANOVA table for regression
anova_table <- anova(lion_model)
anova_table
```

Analysis of Variance Table

```
Response: age_years
      Df Sum Sq Mean Sq F value    Pr(>F)
proportion_black  1 138.544  138.544    49.75 7.677e-08 ***
Residuals       30  83.543    2.785
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Calculate sums of squares manually to understand partitioning
ss_total <- sum((l_df$age_years - mean(l_df$age_years))^2)
ss_residual <- sum(residuals(lion_model)^2)
ss_regression <- ss_total - ss_residual

print("Manual calculation of sums of squares:")
```

```
[1] "Manual calculation of sums of squares:"
```

```
print(paste("SS Total:", round(ss_total, 2)))
```

```
[1] "SS Total: 222.09"
```

```
print(paste("SS Regression:", round(ss_regression, 2)))
```

```
[1] "SS Regression: 138.54"
```

```
print(paste("SS Residual:", round(ss_residual, 2)))
```

```
[1] "SS Residual: 83.54"
```

```
print(paste("SS Regression + SS Residual:", round(ss_regression + ss_residual, 2)))
```

```
[1] "SS Regression + SS Residual: 222.09"
```

Visualizing Variance Components

```
# Create a plot showing variance components
# Get predicted values
l_df$predicted <- predict(lion_model)
mean_age <- mean(l_df$age_years)

# Select one point to illustrate
example_point <- 10

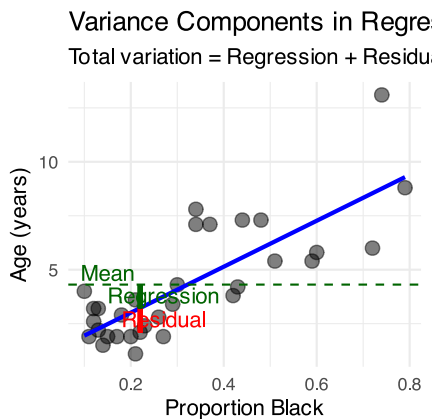
# Create the visualization
variance_plot <- ggplot(l_df, aes(x = proportion_black, y = age_years)) +
  geom_point(size = 3, alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE, color = "blue", size = 1) +
  geom_hline(yintercept = mean_age, linetype = "dashed", color = "darkgreen") +
  # Add lines for one example point
  geom_segment(aes(x = proportion_black[example_point],
                  y = age_years[example_point],
                  xend = proportion_black[example_point],
                  yend = predicted[example_point],
                  color = "red", size = 1) +
  geom_segment(aes(x = proportion_black[example_point],
                  y = predicted[example_point],
```

```

        xend = proportion_black[example_point],
        yend = mean_age),
        color = "darkgreen", size = 1) +
# Add labels
annotate("text", x = 0.15, y = mean_age + 0.5,
        label = "Mean", color = "darkgreen") +
annotate("text", x = l_df$proportion_black[example_point] + 0.05,
        y = (l_df$age_years[example_point] + l_df$predicted[example_point])/2,
        label = "Residual", color = "red") +
annotate("text", x = l_df$proportion_black[example_point] + 0.05,
        y = (l_df$predicted[example_point] + mean_age)/2,
        label = "Regression", color = "darkgreen") +
labs(title = "Variance Components in Regression",
     subtitle = "Total variation = Regression + Residual",
     x = "Proportion Black", y = "Age (years)") +
theme_minimal()

```

variance_plot



Part 6: Comparing Multiple Datasets

Let's practice regression with the prairie biodiversity data:

```

# Fit regression for prairie data
prairie_model <- lm(log_stability ~ species_number, data = p_df)

# Get summary
summary(prairie_model)

```

Call:
lm(formula = log_stability ~ species_number, data = p_df)

Residuals:

Min	1Q	Median	3Q	Max
-0.8146	-0.2165	-0.0094	0.2228	0.7780

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.222902	0.039094	31.281	< 2e-16 ***
species_number	0.028881	0.004694	6.153	5.94e-09 ***

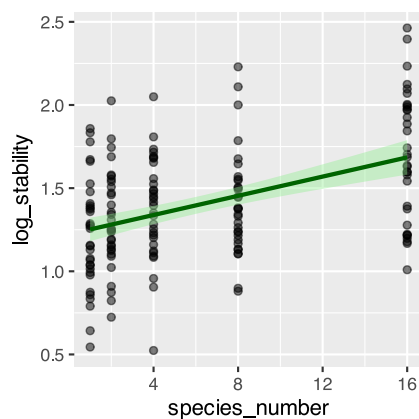
```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3271 on 159 degrees of freedom
Multiple R-squared:  0.1923,    Adjusted R-squared:  0.1872 
F-statistic: 37.86 on 1 and 159 DF,  p-value: 5.94e-09
```

```
# Create plot
prairie_plot <- ggplot(p_df, aes(x = species_number, y = log_stability)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = TRUE, color = "darkgreen", fill = "lightgreen")

prairie_plot
```

```
`geom_smooth()` using formula = 'y ~ x'
```



! Activity 4: Compare the Two Regressions

Compare the lion and prairie regression models:

1. **Which model explains more variance?** (Compare R^2 values)
 - Your answer: _____
2. **Which has a stronger relationship?** (Compare standardized slopes or correlation)
 - Your answer: _____
3. **Which has more precise estimates?** (Compare standard errors relative to estimates)
 - Your answer: _____

Summary and Key Takeaways

What We Learned Today

1. **Correlation vs. Regression:**
 - Correlation: Measures association between two variables
 - Regression: Predicts one variable from another
2. **Assumptions Matter:**
 - Always check assumptions before interpreting results
 - Use appropriate alternatives when assumptions are violated
3. **Interpretation:**
 - R^2 tells us proportion of variance explained
 - Slopes tell us rate of change
 - P-values tell us if relationships are statistically significant
4. **Practical Considerations:**
 - Correlation $\neq$ Causation
 - Outliers can have major impacts
 - Sample size affects power to detect relationships

Common Mistakes to Avoid

1. **Using correlation when you mean regression** (or vice versa)
2. **Ignoring assumption violations**
3. **Extrapolating beyond the range of data**
4. **Confusing confidence and prediction intervals**
5. **Over-interpreting R^2 values**
6. **Forgetting about biological significance vs. statistical significance**

Additional Resources

- Whitlock & Schluter Chapter 16 (Correlation)
- Whitlock & Schluter Chapter 17 (Regression)
- R for Data Science: <https://r4ds.had.co.nz/>
- Quick-R Regression: <https://www.statmethods.net/stats/regression.html>